

## Exercise 13

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### Question 1

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Ekman velocity - Zonal & Meridional  
Components

$$\frac{\partial u_E}{\partial t} - f v_E = A_v \frac{\partial^2 u_E}{\partial z^2}$$

$$\frac{\partial v_E}{\partial t} + f u_E = A_v \frac{\partial^2 v_E}{\partial z^2}$$

Grid starts at  $z_0 = 0$   
 $z_k = z_0 - k \Delta z$

$$t_i = t_0 + n \Delta t$$

$$\frac{\partial u_E}{\partial t} = \frac{u_k^{n+1} - u_k^n}{\Delta t}$$

$$\frac{\partial v_E}{\partial t} = \frac{v_k^{n+1} - v_k^n}{\Delta t}$$

Time derivative  
- Forward difference

$$\frac{\partial^2 u}{\partial z^2} = \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta z^2}$$

$$\frac{\partial^2 v}{\partial z^2} = \frac{v_{i+1}^n - 2v_i^n + v_{i-1}^n}{\Delta z^2}$$

Vertical diffusion terms  
- centered difference approximation

### Discretised Equations

$$\frac{u_k^{n+1} - u_k^n}{\Delta t} - f v_k^n = A_v \left( \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta z^2} \right)$$

$$\Rightarrow u_k^{n+1} = u_k^n + f v_k^n \Delta t + \frac{A_v \Delta t}{\Delta z^2} (u_{i+1}^n - 2u_i^n + u_{i-1}^n) \quad \left. \vphantom{\frac{A_v \Delta t}{\Delta z^2}} \right] \text{Zonal}$$

$$\frac{v_k^{n+1} - v_k^n}{\Delta t} + f u_k^n = A_v \left( \frac{v_{i+1}^n - 2v_i^n + v_{i-1}^n}{\Delta z^2} \right)$$

$$\Rightarrow v_k^{n+1} = v_k^n - f u_k^n \Delta t + \frac{A_v \Delta t}{\Delta z^2} (v_{i+1}^n - 2v_i^n + v_{i-1}^n) \quad \left. \vphantom{\frac{A_v \Delta t}{\Delta z^2}} \right] \text{Meridional}$$

## Question 2

Stability comes from the diffusion term  
→ ignore coriolis

$$\frac{\partial u}{\partial t} = A_v \frac{\partial^2 u}{\partial z^2}$$

$$\rightarrow u_k^{n+1} = u_k^n + \frac{A_v \Delta t}{\Delta z^2} (u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

$$D = \frac{A_v \Delta t}{\Delta z^2} \rightarrow \text{diffusive Courant Number}$$

For von Neumann stability:  $D \leq \frac{1}{2}$

$$\therefore \frac{A_v \Delta t}{\Delta z^2} \leq \frac{1}{2}$$

### Question 3

$$\frac{A_v \Delta t}{\Delta z^2} \leq \frac{1}{2} \quad \text{or} \quad \Delta t \leq \frac{\Delta z^2}{2 A_v}$$

$$A_v = 10^{-4} \text{ m}^2 \text{ s}^{-1} \Rightarrow \Delta t \leq \frac{\Delta z^2}{2 \times 10^{-4}}$$

• If  $\Delta z = 1$

$$\Delta t \leq \frac{1^2}{2 \times 10^{-4}}$$

$$\Delta t \leq 5000 \text{ s} \quad \text{so } \Delta t = 3600 \text{ s (1 hr) works}$$

• If  $\Delta z = 0.5$

$$\Delta t \leq \frac{0.5^2}{2 \times 10^{-4}}$$

$$\Delta t \leq 1250 \text{ s} \rightarrow \text{could use } \Delta t \sim 1000 \text{ s}$$

• If  $\Delta z = 10 \text{ m}$

$$\Delta t \leq \frac{10^2}{2 \times 10^{-4}}$$

$$\Delta t \leq 500000 \text{ s} \quad (\sim 5.8 \text{ days})$$

→ Stable but resolution is too coarse

∴ I would recommend  $\Delta z = 1 \text{ m}$ ,  $\Delta t = 3600 \text{ s}$

This is stable and gives a high enough resolution to resolve the Ekman layer.

$$D = \frac{A_v \Delta t}{\Delta z^2} = \frac{10^{-4} \times 3600}{1^2} = 0.36 \leq \frac{1}{2}$$